

Assignment2

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data introduction

Data was collected from a class of 48 male and female students. Several questions were asked, such as major, minor, height, and weight. To conduct analysis and develop a scientific project to be discussed in class and used for graduation, the data was entered into Excel and cleaned. The types of variables used were qualitative (such as major, gender, minor, etc.), quantitative (such as weight, height, and cumulative GPA), Data were collected using papers distributed to male and female students.

Before conducting data analysis, the variables were classified into qualitative and quantitative to conduct the appropriate analysis for each type and what are the appropriate statistical tests.

data collection tool

- Papers distributed between males and females

data cleaning

*The “Year at University” column has been changed from text to numbers.

*The (minor) column has been filled with the word (none) in place of the empty cells.

*Empty cells were considered missing data

*The column (Do you have a laptop or desktop) has been converted from text values (laptop, desktop) to values (yes,no) to make the data more accurate.

- The length variable was converted (to unitless values).
- read data

```
data3=read_excel("C:AL SAFA.xlsx",sheet = 1)
```

```
library(readxl)
```

```
## Warning: package 'readxl' was built under R version 4.4.3
```

```

Data <- read_excel("~/Data.xlsx")
View(Data)
print(Data)

## # A tibble: 48 × 12
##   GENDER MAJOR      MINOR `year in university` GPA `grade in stat
101`
##   <chr>  <chr>      <chr>                <dbl> <dbl>                <
dbl>
## 1 FEMALE statistic NONE                4  69.6
61
## 2 FEMALE statistic NONE                5  67
77
## 3 FEMALE statistic NONE                5  70
75
## 4 FEMALE statistic NONE                6  60.8
50
## 5 FEMALE statistic NONE                4  64.3
72
## 6 FEMALE statistic NONE                5  69.3
83
## 7 FEMALE statistic NONE                5  67.2
75
## 8 MALE   statistic NONE                5  69.1
80
## 9 MALE   statistic NONE                6  61
50
## 10 FEMALE statistic NONE                5  66.8
50
## # i 38 more rows
## # i 6 more variables: `grade in stat 201` <dbl>, `did you use r befo
re` <chr>,
## #   `do you have laptop or desktop at home` <chr>, weight <dbl>, hei
ght <dbl>,
## #   `blood type` <chr>

```

#Descriptive statistic

```

summary(Data)

##   GENDER      MAJOR      MINOR      year in un
iversity
## Length:48      Length:48      Length:48      Min.   :3.
000
## Class :character Class :character Class :character 1st Qu.:4.
000
## Mode  :character Mode  :character Mode  :character Median :4.
000
##                                     Mean   :4.
542
##                                     3rd Qu.:5.

```

```

000
##                                     Max.      :7.
000
##
##      GPA      grade in stat 101 grade in stat 201 did you use r b
efore
## Min.      :60.00   Min.      :50.00   Min.      :50.00   Length:48
## 1st Qu.:65.00   1st Qu.:60.00   1st Qu.:63.00   Class :characte
r
## Median :67.30   Median :70.00   Median :70.00   Mode  :characte
r
## Mean    :68.83   Mean    :67.04   Mean    :68.76
## 3rd Qu.:70.00   3rd Qu.:75.00   3rd Qu.:75.00
## Max.    :84.20   Max.    :89.00   Max.    :90.00
##                                     NA's      :1      NA's      :19
## do you have laptop or desktop at home      weight      height
## Length:48                                     Min.      : 42.00   Min.      : 1.
64
## Class :character                                     1st Qu.: 57.50   1st Qu.:158.
00
## Mode  :character                                     Median : 63.50   Median :163.
00
##                                     Mean    : 67.12   Mean    :163.
25
##                                     3rd Qu.: 72.50   3rd Qu.:176.
25
##                                     Max.    :110.00   Max.    :189.
00
##
##      blood type
## Length:48
## Class :character
## Mode  :character
##
##
##
##

```